JOAQUIN LEAL ENCISO

Beijing, China

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RESEARCH INTERESTS

Operations Research & Decision Analysis · Transportation System Optimization · Logistics & Supply Chain Optimization · Machine Learning

EDUCATION

Beihang University (BUAA) — *Beijing, China*
M.S. in Management Science and Engineering — Sept 2024 – Jun 2027

- Thesis: "Optimization of Charging Scheduling for Battery Electric Buses (BEBs) Considering SoC-Dependent Charging Power." Advisor: Prof. WANG Heixin.
- GPA: 3.68/4.00. Relevant coursework: Operations Research, Supply Chain Management, Macroeconomics and Microeconomics.

Industrial University of Santander — *Bucaramanga, Colombia*
B.Eng. in Mechanical Engineering — Sept 2011 – Jun 2016

- Relevant coursework: Programming, Finite Element Analysis, and Economic Engineering.

RESEARCH EXPERIENCE

Beihang University — *Beijing, China*
Graduate Research Student — Sept 2025 – Present

- Developed a **Mixed-Integer Linear Programming (MILP)** model for battery-electric bus charging and scheduling to minimize energy and operational costs under realistic fleet and infrastructure constraints.
- Implemented an optimization framework in **Python** using **Gurobi Optimizer** on a real-world battery-electric bus fleet case study a major BRT system in Bogotá, Colombia.
- Conducted a systematic review of more than 40 studies, identifying research gaps related to SoC-dependent charging-power modeling, electricity tariffs, and operational constraints.

Energy Lab — *Bucaramanga, Colombia*
Undergraduate Research Assistant — 2016

- Performed finite element analysis (FEA) to characterize the mechanical behavior and structural integrity of Fiberspar composite pipes for crude-oil transportation applications in Colombia.
- Contributed to the mechanical characterization of a composite pipe technology newly introduced into the Colombian oil-transportation sector.

PUBLICATIONS

- **González-Estrada, O. A., Leal Enciso, J., & Reyes-Herrera, J. D.** (2016). "Structural integrity analysis of composite pipes for the transport of crude oil by finite elements." *UIS Ingenierías*, *15*(2), 105–106. DOI: 10.18273/revuin.v15n2-2016009

TECHNICAL SKILLS

Programming:	Python, LaTeX
Optimization:	Gurobi Optimizer, Mixed-Integer Linear Programming (MILP), Mathematical Programming, Scheduling
AI & Data-Driven Methods:	AI-Assisted Decision-Making
Analysis:	Sensitivity Analysis, Decision Analysis, Optimization Modeling
Software:	VS Code, Jupyter, Plant Simulation
Languages:	Spanish (Native), English (TOEFL), Mandarin Chinese (HSK 4)

PROFESSIONAL EXPERIENCE

Wärtsilä *Bogotá, Colombia*
Spare Parts Coordinator Aug 2023 – Sept 2024

- Coordinated quotations, order processing, and international logistics for maritime spare parts and critical engine components.
- Served as the coordination link between customers and internal departments including Field Service, Supply Chain, and Finance, supporting the resolution of technical and commercial non-conformities.
- Monitored response times and supply-chain flows to support service quality and timely delivery of critical vessel-engine components.

Contransmagdalena – Regional Transport Company *Bucaramanga, Colombia*
Maintenance Mechanical Engineer Feb 2018 – Aug 2021

- Developed a maintenance program for a fleet of diesel buses, supporting compliance with safety and service-level requirements.
- Organized and digitized technical vehicle records, improving data accessibility and maintenance-information management.
- Supported transportation-route and scheduling planning based on passenger demand, contributing to fleet utilization and service reliability.

AWARDS & HONORS

Master's Degree Full Scholarship, Chinese Scholarship Council *2024*

ADDITIONAL TRAINING

Smart Cities Summer Course, Zhejiang University *2025*
Focus areas: Smart Mobility, Urban Planning, and Sustainable Cities.

Supply Chain Logistics, Rutgers University (Coursera) *2024*

Python for Everybody, University of Michigan (Coursera) *2024*